

# Behrouz Akhbari

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## EDUCATION

### University of Toronto

Honours Bachelor of Science, Double Major in Computer Science and Statistics

Toronto, ON

Expected Graduation: 2025

### Certifications

AWS Certified Cloud Practitioner

## EXPERIENCE

### Software Engineering Intern | *React, Express/Node, SPARQL, AWS*

Jun 2023 – Present

Urban Research Data Center, University of Toronto

Toronto, ON

- Developed **REST APIs** using **Express** and **SPARQL** to efficiently query data from a knowledge graph, reducing data retrieval time by 30%, optimizing data retrieval processes.
- Implemented new functionality** in a web application by leveraging our existing knowledge graph data, **enhancing data accessibility** and **improving user interaction** with the system.
- Utilized **React (MUI)** to build a dynamic frontend allowing users to input data, enhancing the data query process.
- Implemented React for **data visualization, generating interactive graphs, maps, and tables** for comprehensive representation of acquired data.
- Deployed and hosted the application on **AWS** using **Elastic Beanstalk** to ensure scalability, performance and security.
- Implemented a **CI/CD pipeline** using **AWS CodePipeline** and **CodeBuild** to automate deployments.

### Front-End Intern | *React, Flask*

Jan 2023 – Aug 2023

Gene2Lead

Mississauga, ON

- Developed a **responsive and interactive website** using **React** for the front end.
- Utilized **Flask** query results to dynamically populate and manage data tables in the front end.

## PROJECTS

### Easy-Chef (Recipe Social Media App) | *React, Bootstrap, Django, REST, SQLite, GitLab*

- Developed a **dynamic web application** where users can **create, share, and discover recipes**.
- Implemented a robust search functionality that allows users to **find recipes based on tags or keywords**.
- Designed and **optimized the database schema**, ensuring efficient data storage and retrieval for recipes and user interactions.
- Employed **JWT token authentication** and serialization techniques to secure user accounts and **protect sensitive information**.

### TherapyMatch – Hackathon | *React, MUI, Flask, OpenAI API, OpenStreetMap API, SQLite, GitHub*

- A web application that connects users with therapists based on their unique needs and location.
- Utilized the **OpenAI GPT-3.5 model** to analyze user input and identify key points related to their mental health needs.
- Integrated **OpenStreetMap API** to facilitate geocoding and distance calculations.
- Hosted the application on **Microsoft Azure** for scalable and reliable performance.

### 6502 CPU Emulation | *C++, Assembly*

- Developed an accurate **instruction set simulation**, ensuring precise execution of opcodes and instructions
- Implemented comprehensive memory management system for accurate emulation of the 6502 CPU's **addressing modes, including zero page, absolute, indirect, and indexed addressing**.
- Debugged and tested** the emulator using **GDB** and test programs, ensuring compatibility and correctness with original hardware behaviour.

## TECHNICAL SKILLS

**Languages:** Java, Python, C/C++, JavaScript, HTML/CSS, Assembly (RISC-V, 6502)

**Frameworks:** React, NodeJS, ExpressJS, NextJS, Flask, Django, MUI

**Other:** SQL (SQLite, Postgres), GraphDB / SPARQL, MongoDB (Mongoose), Git, Figma, OpenAI API, Postman, pandas, NumPy, Matplotlib, TensorFlow, PyTorch, Bootstrap, jQuery, AWS